

ARTHUR PEREIRA VIEGAS

Full-Stack & AI Engineer

Porto Alegre, Brazil · Open to remote / Ireland

arthurviegas@gmail.com · github.com/ArthurViegas01 · linkedin.com/in/arthurpviegas · arthurviegas.netlify.app

SUMMARY

Full-Stack & AI Engineer with 4 years of experience building and scaling data-intensive web applications and production AI systems. Specialized in end-to-end RAG pipelines, multi-agent orchestration with LangGraph, MCP servers, and AWS cloud architecture. At Dataglass, led the full Dockerization and CI/CD automation of a production platform serving healthcare clients via Tableau, eliminating 100% of manual deployment steps. Currently shipping a multi-tenant WhatsApp AI SaaS (Encaixe) built with Next.js, FastAPI, and LangGraph, with PostgreSQL row-level-security tenancy and production-grade observability. Seeking mid-to-senior IC roles in product-oriented engineering teams (Ireland / remote).

TECHNICAL SKILLS

AI / LLM Engineering: RAG pipelines (ingestion → chunking → embedding → retrieval → grounded generation), Hybrid Search (vector + keyword) with Reciprocal Rank Fusion, LangGraph (multi-node agents, conditional routing, durable Postgres checkpointing), LangChain, MCP / Model Context Protocol servers (FastMCP), token-by-token LLM streaming, multi-provider LLM abstraction (Claude, Groq/Llama 3, Gemini/Vertex AI, OpenAI, Ollama), function/tool calling, semantic search, pgvector (HNSW), embeddings (Voyage AI, Sentence Transformers / fastembed-ONNX), prompt engineering

Languages: Python, TypeScript, JavaScript (ES2022+), SQL, HTML5, CSS3 / Tailwind / SASS

Back-End: FastAPI, Django, Django REST Framework, Celery, Node.js, async Python (asyncio, asyncpg), SQLAlchemy 2.0, Pydantic v2, REST APIs, microservices, ASGI middleware, idempotent webhooks

Front-End: React 18, Next.js (14/15, App Router), Vue 3 (Composition API, Pinia), TypeScript, Vite, Tailwind CSS, state management (Zustand, Pinia, Context API + useReducer), Monaco Editor, Recharts, Framer Motion, i18n, responsive design, accessibility (ARIA)

Cloud & DevOps: AWS (CodePipeline, CodeBuild, Elastic Beanstalk, EC2, S3, ECR, RDS, ElastiCache, Secrets Manager), Google Cloud, Docker & Docker Compose, Terraform / IaC (Railway, Netlify, Hetzner, AWS providers), CI/CD (GitHub Actions), GHCR, Nginx, Railway, Vercel, Netlify

Databases: PostgreSQL, pgvector, Redis / ElastiCache / Upstash, Supabase, Row-Level Security (multi-tenancy), Alembic migrations

Observability & Reliability: Sentry, structlog, health checks, rate limiting, circuit breakers, retry/backoff (tenacity), request-ID tracing

Testing & Code Quality: pytest (asyncio, coverage), respx, fakeredis, Vitest / Testing Library, Ruff, mypy (strict), ESLint, integration testing (tenant isolation), TDD

Architecture & Practices: Multi-tenant SaaS, event-driven systems, ports & adapters, layered/stateless services, Git / Git Flow, monorepos, Agile / Scrum / Kanban, code review & mentoring, LGPD/data-privacy compliance

Data & Analytics: Tableau (API integration), data visualization, data migration, document processing (PDF/DOCX), Portuguese NLP

Spoken Languages: Portuguese (Native), English (Advanced / Professional)

PROFESSIONAL EXPERIENCE

Full-Stack Developer · Dataglass

Jan 2024 – Present

- **Technical leadership:** led development of the company's main data-visualization platform (Django + React) serving healthcare clients via Tableau Cloud APIs; mentor a junior intern through structured code reviews and sprint planning, reducing PR review cycles by ~40%.
- **DevOps & Cloud:** architected and implemented full Dockerization of the Django application, managing images via AWS ECR and cutting environment setup from 2+ hours to under 5 minutes.
- **CI/CD automation:** built a fully automated pipeline with AWS CodePipeline and CodeBuild deploying to Elastic Beanstalk, eliminating 100% of manual deployment steps and human-error incidents.
- **Cloud migration:** led migration of application data and assets to AWS S3, decoupling file storage from the server and improving scalability and availability.

Full-Stack Development Intern · Dataglass

May 2023 – Dec 2023

- Progressed rapidly from front-end (React) tasks to full-stack responsibilities on the core Django application, earning an early promotion within 7 months.
- Developed and maintained data-analysis and visualization features used daily by healthcare clients, consistently delivering Kanban sprint tasks ahead of schedule.

Community Analysis Intern · Dell Technologies / PUC-RS

Apr 2021 – May 2023

- Collaborated with a globally distributed team (USA, India) to moderate the official Dell Brazil community forum; authored technical guides that proactively reduced repetitive support queries, and delivered weekly data reports to international management.

SELECTED PROJECTS

Encaixe — Multi-Tenant WhatsApp AI SaaS

FastAPI · LangGraph · Claude Haiku · Voyage AI · pgvector · Celery · Next.js 14 · Supabase · Terraform

Customer-service SaaS where a 6-node LangGraph agent classifies intent, answers FAQs via Voyage AI + pgvector semantic search, books appointments through Google Calendar OAuth2, and escalates to a human on low confidence. Tenant isolation via PostgreSQL row-level security; production-grade with Sentry/structlog observability, Redis rate limiting + circuit breaker, idempotent webhooks, and GitHub Actions CI/CD to a Hetzner VPS and Vercel.

Context — RAG Pipeline over Private Documents

FastAPI · Celery · pgvector · fastembed/ONNX · LangChain · React · Docker · Nginx

RAG system over user-uploaded PDFs/DOCX: an async Celery pipeline parses, chunks, and embeds (fastembed/ONNX) into pgvector, with hybrid retrieval (vector + Portuguese full-text search) fused via Reciprocal Rank Fusion. Multi-provider LLM (Groq/Ollama/OpenAI), containerized for Railway or a self-hosted VPS behind Nginx, with CI and coverage.

Devscope — MCP Server for AI Agents

Python · FastMCP · GitHub API · Groq/Llama 3.1-70b · LangChain · Docker · Terraform

Model Context Protocol server exposing four GitHub profile-analysis tools to LLM clients, with token-by-token streaming over Streamable HTTP and structured JSON output via LangChain. Redis caching and a Lua-based atomic rate limiter; multi-stage Docker provisioned with Terraform (Railway + Netlify, region-policy validation) and published to GHCR.

Synth — Real-Time AI Component Generator

Next.js 15 · multi-provider LLM · Monaco Editor · Babel Standalone · Zustand

Streams LLM-generated React/TSX token-by-token into a Monaco editor and compiles it live in the browser with Babel Standalone inside a hardened iframe sandbox. A single Next.js streaming route abstracts four LLM providers; includes client-side validation and an automatic mock-props generator inferred from TypeScript types.

EDUCATION

B.S. in Software Engineering · PUC-RS, Porto Alegre

2021 – 2025

- Built production websites and applications for external companies from the first year onward through PUC-RS's industry-partnership program, accumulating hands-on software-development experience throughout the degree.